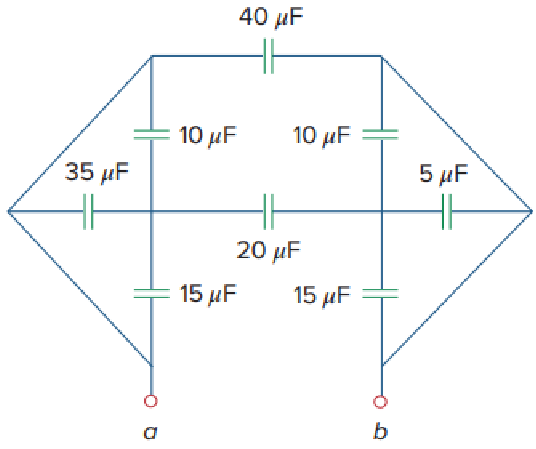
Chapter 6, Problem 22P

Problem

Obtain the equivalent capacitance of the circuit in Fig. 6.56.

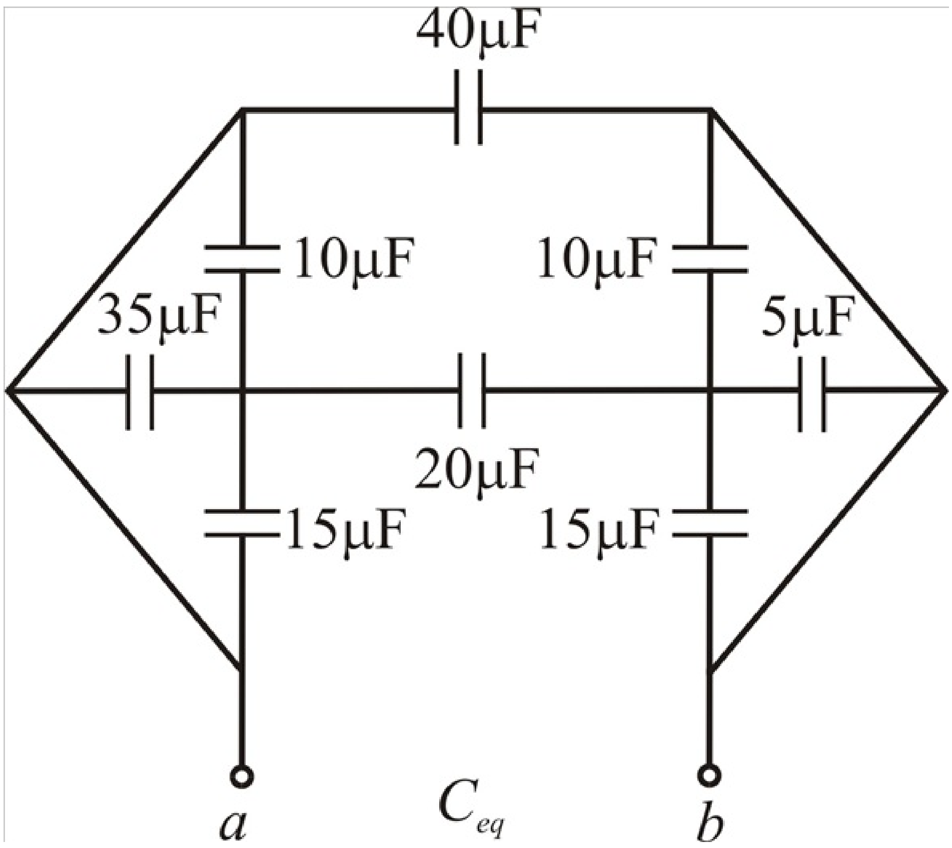
Fig. 6.56:



Step-by-step solution

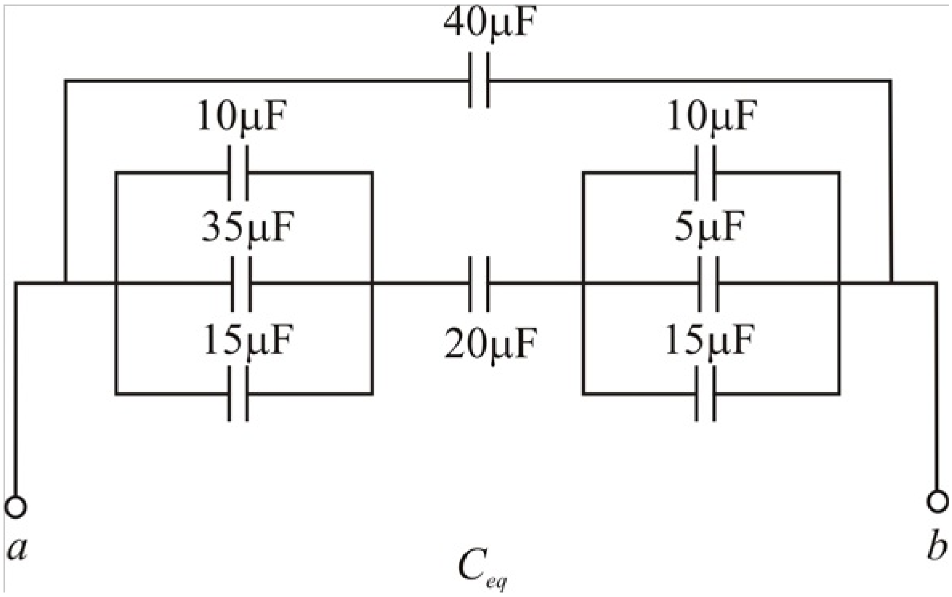
Step 1 of 6

Given circuit diagram



Step 2 of 6

Above circuit can be represented as below



Step 3 of 6

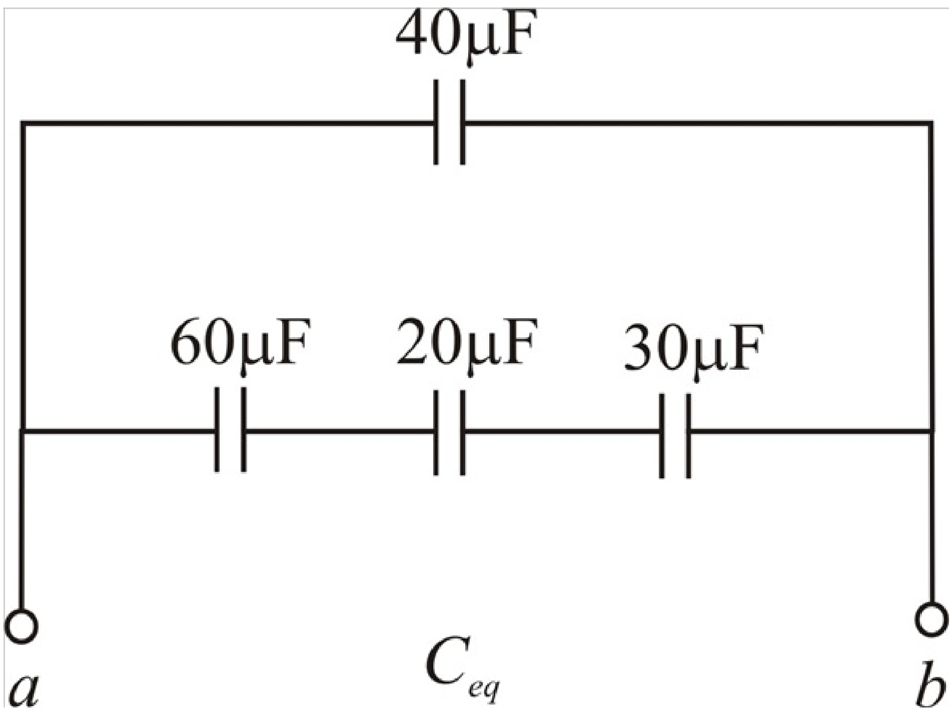
In the above circuit $10\mu F, 35\mu F$ and $15\mu F$ are in parallel

$$10\mu F + 35\mu F + 15\mu F = 60\mu F$$

In the above circuit $10\mu F, 5\mu F$ and $15\mu F$ are in parallel

$$10\mu F + 5\mu F + 15\mu F = 30\mu F$$

Now above circuit reduced to



Step 4 of 6

In the above circuit $60\mu F, 20\mu F$ and $30\mu F$ are in series, so their equivalent capacitance $C_{eq}^{'}$ is

$$C_{eq}^{'} = \frac{1}{\frac{1}{60\mu} + \frac{1}{20\mu} + \frac{1}{30\mu}}$$

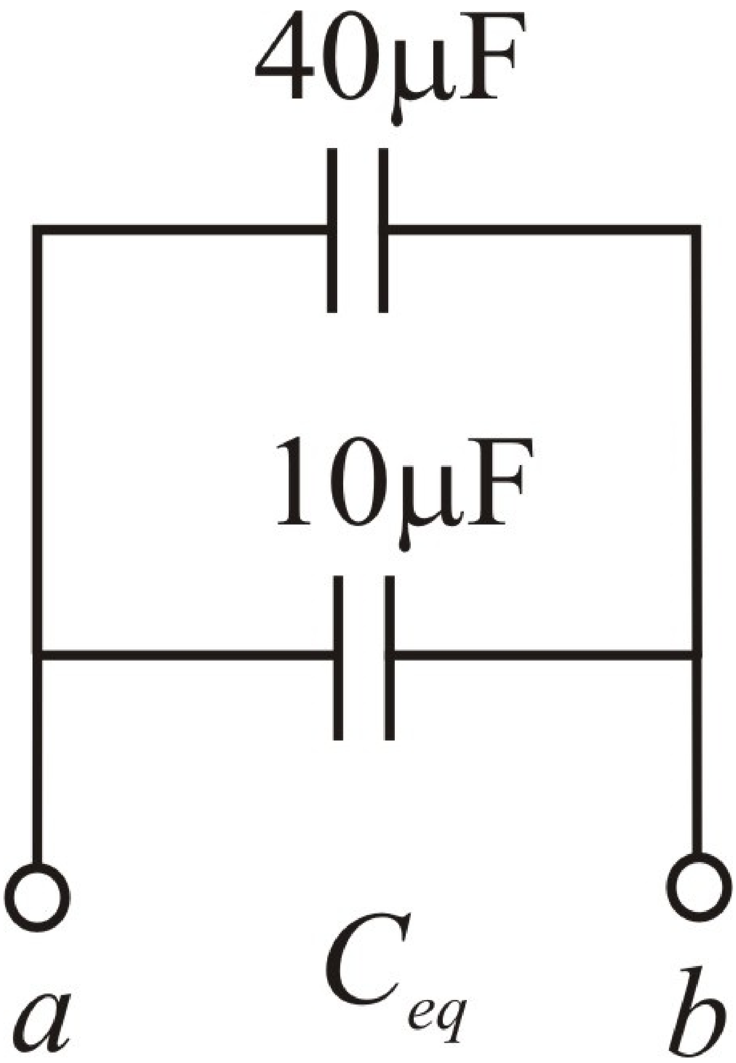
$$C_{eq}^{'} = \frac{1}{\frac{1+3+2}{60\mu}}$$

$$C_{eq}^{'} = \frac{60\mu}{6}$$

$$C_{eq}^{'} = 10\mu$$

Step 5 of 6

Above circuit becomes



Step 6 of 6

In the above circuit $40\mu F$ and $10\mu F$ are in parallel

$$C_{eq} = 40\mu F + 10\mu F$$

$$\boxed{C_{eq} = 50\mu F}$$